01 MINIMAL NOTES

Complex numbers: only the machinery you use

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 5–22

Margin references. "PS1 q3" means Problem Sheet 1, Question 3. Each reference sits beside the theory or method most relevant to that exercise.

Purpose. This is deliberately compressed: definitions, structural facts and reusable formulas only. Long computations and most proofs are omitted.

Basic form, conjugate and modulus. [0.1.1–0.1.9] Every $z \in \mathbb{C}$ is $z = a + ib$ with $i^2 = -1$. Then

$$\Re z = a, \qquad \Im z = b, \qquad \bar{z} = a - ib, \qquad |z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}.$$

Useful identities:

$$\overline{z + w} = \bar{z} + \bar{w}, \quad \overline{zw} = \bar{z}\bar{w}, \quad |zw| = |z||w|, \quad \frac{1}{z} = \frac{\bar{z}}{|z|^2}, \quad |z + w| \le |z| + |w|.$$

Geometrically $z = a + ib$ is the point (a, b); $|z - w|$ is Euclidean distance, and conjugation reflects in the real axis.

Argument and polar form. [0.2.2–0.2.5] For $z \neq 0$,

$$z = r(\cos\theta + i\sin\theta) = re^{i\theta}, \qquad r = |z|.$$

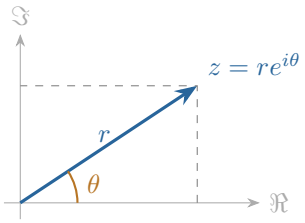
Arguments differ by $2\pi k$. The unique principal argument is $\operatorname{Arg} z \in (-\pi, \pi]$. Euler's formula is

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}, \qquad e^{i\pi} + 1 = 0.$$

Hence, for $z = re^{i\theta}$ and $w = \rho e^{i\phi}$,

$$zw = r\rho e^{i(\theta+\phi)}, \qquad \frac{z}{w} = \frac{r}{\rho}e^{i(\theta-\phi)}.$$

So multiplication = scale moduli and add angles.



De Moivre and roots. [0.2.7; 0.3.1–0.3.2]

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta), \qquad (re^{i\theta})^n = r^n e^{in\theta}.$$

The nth roots of unity are

$$\boxed{e^{2\pi ik/n}, \quad k = 0, \dots, n-1,}$$

the vertices of a regular n-gon. If $w = \rho e^{i\phi} \neq 0$, all solutions of $z^n = w$ are

$$\boxed{z_k = \rho^{1/n} e^{i(\phi+2\pi k)/n}, \quad k = 0, \dots, n-1.}$$

Complex logarithm and powers. [0.3.3] $e^u = z$ has infinitely many solutions because $e^{2\pi i} = 1$:

$$\log z = \log|z| + i(\arg z + 2\pi k), \quad k \in \mathbb{Z}.$$

The principal logarithm is $\operatorname{Log} z = \log|z| + i\operatorname{Arg} z$. Complex powers are defined by $z^w = e^{w\log z}$ and are generally multi-valued.

02 MINIMAL NOTES

Vectors, geometry and matrices

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 23–40

Linear combinations and span. [1.2.1–1.2.4] For $v_1, \dots, v_m \in \mathbb{R}^n$ a linear combination is $\sum_k \alpha_k v_k$. Their span is PS1 q2–3

$$\operatorname{span}\{v_1, \dots, v_m\} = \left\{ \sum_{k=1}^{m} \alpha_k v_k : \alpha_k \in \mathbb{R} \right\}.$$

For a nonzero v, $\operatorname{span}\{v\}$ is the line through 0 in direction v. Two non-parallel vectors span a plane through 0.

Norm, distance and dot product. [1.3.1–1.3.8] For $u, v \in \mathbb{R}^n$, PS1 q4,7

$$\|v\| = \sqrt{\sum v_k^2}, \qquad d(u,v) = \|u - v\|, \qquad u \cdot v = \sum u_k v_k.$$

Core facts:

$$\|\alpha v\| = |\alpha| \, \|v\|, \qquad \|u + v\| \le \|u\| + \|v\|, \qquad |u \cdot v| \le \|u\| \, \|v\|.$$

In $\mathbb{R}^2, \mathbb{R}^3$,

$$\boxed{u \cdot v = \|u\| \, \|v\| \cos\theta}; \qquad u \perp v \iff u \cdot v = 0.$$

Lines and planes. [1.3.1] A line through v_0 parallel to $p \neq 0$ is PS1 q5–6

$$v = v_0 + tp, \qquad t \in \mathbb{R}.$$

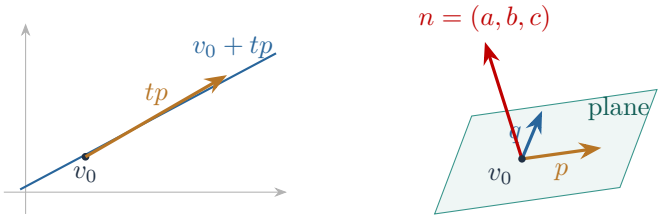
A plane through v_0 parallel to independent p, q is

$$v = v_0 + tp + sq, \qquad s, t \in \mathbb{R}.$$

In $\mathbb{R}^3$, a plane also has Cartesian form

$$av_1 + bv_2 + cv_3 = d,$$

where $n = (a, b, c)$ is a normal vector, perpendicular to both directions p, q, so $n \cdot p = n \cdot q = 0$.



Complex vectors. [1.4] For $z \in \mathbb{C}^n$ use

$$\|z\| = \sqrt{\sum |z_k|^2}, \qquad z \cdot w = \sum z_k \overline{w_k}.$$

The conjugation is essential: it makes $z \cdot z = \|z\|^2 \ge 0$.

Matrices. [1.5] An $m \times n$ matrix has entries a_{ij}. The transpose satisfies $(A^T)^T = A$. Matrix multiplication is defined only when inner dimensions agree: PS2 q1; PS6 q5

$$(AB)_{ij} = \sum_k a_{ik} b_{kj} = (\text{row } i \text{ of } A) \cdot (\text{column } j \text{ of } B).$$

Row-by-column rule: entry (i, j) is the dot product of row i from A with column j from B. For A of size $m \times n$ and B of size $n \times p$, the answer has size $m \times p$. Usually $AB \neq BA$.

02 WORKED METHODS

Span, row-by-column multiplication and planes

Small integer examples; the method is the point.

Checking whether a vector belongs to a span

PS1 q3

Let $u = (1, 1, 0)$ and $v = (0, 1, 1)$. Does $b = (2, 3, 1)$ belong to $\text{span}\{u, v\}$?

$$\alpha u + \beta v = b \iff (\alpha, \alpha + \beta, \beta) = (2, 3, 1) \iff \alpha = 2,\ \beta = 1.$$

Yes: $b = 2u + v$. But $c = (1, 0, 1)$ is **not** in the span: the first and last coordinates force $\alpha = 1$, $\beta = 1$, whereas the middle would be $2 \neq 0$. *Always compare all coordinates.*

Matrix multiplication: rows against columns

PS2 q1(b); PS6 q5

$$A = \begin{pmatrix} 1 & 2 & 0 \\ -1 & 1 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 2 \end{pmatrix}.$$

The dimensions are $(2 \times 3)(3 \times 2)$, so AB is 2×2.

$$AB = \begin{pmatrix} 1 \cdot 2 + 2 \cdot 0 + 0 \cdot 1 & 1 \cdot 1 + 2(-1) + 0 \cdot 2 \\ (-1)2 + 1 \cdot 0 + 3 \cdot 1 & (-1)1 + 1(-1) + 3 \cdot 2 \end{pmatrix} = \boxed{\begin{pmatrix} 2 & -1 \\ 1 & 4 \end{pmatrix}}.$$

Recipe: fix row i of the first matrix and column j of the second, multiply matching entries, add. The resulting number goes in position (i, j). Never multiply matrices entrywise.

From a normal vector to a plane equation

PS1 q6

A plane passes through $v_0 = (1, 2, 0)$ and has normal $n = (1, -1, 2)$. Every point $v = (x, y, z)$ on it satisfies

$$n \cdot (v - v_0) = 0 \iff (x - 1) - (y - 2) + 2z = 0 \iff \boxed{x - y + 2z = -1}.$$

Reverse direction: the coefficients $(1, -1, 2)$ in the Cartesian equation are the normal. Direction vectors p, q in the plane must satisfy $n \cdot p = n \cdot q = 0$.

03 MINIMAL NOTES

Vector spaces, subspaces, bases and dimension

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 41–58

Vector space. [2.0.1] A vector space V over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ is a set closed under vector addition and scalar multiplication, obeying the usual algebraic laws (associativity, commutativity of addition, zero and inverses, scalar identity/associativity and distributivity). Typical examples: $\mathbb{F}^n$, polynomials P_n, matrices $M_{m,n}(\mathbb{F})$, functions, and solution spaces of homogeneous linear ODEs. PS4 q1

Subspace test. [2.2.1–2.2.3] A nonempty $W \subseteq V$ is a subspace iff PS4 q2–3,6

$$\boxed{u, v \in W,\ \alpha, \beta \in \mathbb{F} \quad \Longrightarrow \quad \alpha u + \beta v \in W.}$$

Consequently every subspace contains 0 and every linear combination of its elements. Also $\operatorname{span}\{v_1, \dots, v_k\}$ is the smallest subspace containing those vectors.

Basis and coordinates. [2.3.1–2.3.6] A list $E = \{e_1, \dots, e_n\}$ is a basis of V iff every $v \in V$ has a *unique* representation PS5 q1–2

$$v = \alpha_1 e_1 + \cdots + \alpha_n e_n.$$

Its coordinate vector is

$$[v]_E = (\alpha_1, \dots, \alpha_n)^T \in \mathbb{F}^n.$$

Thus choosing a basis turns an abstract finite-dimensional vector space into coordinates in $\mathbb{F}^n$.

Spanning versus linear independence. [2.3.7–2.3.17] PS5 q1,3

- $v_1, \dots, v_n$ **span** V iff every $v \in V$ is some linear combination of them.
- They are **linearly independent** iff

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0 \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_n = 0.$$

- They are linearly dependent iff one vector is a linear combination of the others; that vector can be removed without changing the span.

Basis criterion [2.3.17].

$$\boxed{\text{basis} \iff \text{spanning and linearly independent.}}$$

Existence gives spanning; uniqueness is exactly linear independence.

Dimension. [2.3.20–2.3.23] All bases of a finite-dimensional space have the same size; this is $\dim V$. If $\dim V = n$: PS5 q4–6

- fewer than n vectors cannot span V;
- more than n vectors cannot be linearly independent;
- every linearly independent set extends to a basis;
- every finite spanning set contains a basis;
- **exactly n vectors are a basis iff they are spanning iff they are linearly independent.**

Examples: $\dim \mathbb{F}^n = n$, $\dim P_n = n + 1$, $\dim M_{m,n}(\mathbb{F}) = mn$.



with exactly $n = \dim V$ vectors,
all three conditions are equivalent

03 WORKED METHODS

Subspaces, independence and finding a basis

Small integer examples; the method is the point.

The subspace test, in function spaces

PS4 q1–3

Inside P_3 (polynomials of degree at most 3), consider

$$W = \{p \in P_3 : p(0) = p(1)\}, \qquad U = \{p \in P_3 : p(0) = 1\}.$$

W **is a subspace:** the zero polynomial lies in W; if $p, q \in W$ then for any $a, b \in \mathbb{R}$,

$$(ap + bq)(0) = ap(0) + bq(0) = ap(1) + bq(1) = (ap + bq)(1).$$

U **is not:** it does not contain the zero polynomial. Similarly the nonnegative real-valued functions do not form a subspace: multiplication by -1 fails.

Independent or dependent? Do not guess from a picture

PS5 q1,3

Take $u = (1, 0, 1)$ and $v = (0, 1, 1)$. Solve

$$\alpha u + \beta v = (\alpha, \beta, \alpha + \beta) = (0, 0, 0).$$

The first two entries force $\alpha = \beta = 0$, so u, v are **independent**. Now let $w = (1, 1, 2) = u + v$: then

$$u + v - w = 0$$

is a **nontrivial** relation, so u, v, w are **dependent**. Their span is the same as the span of u, v.

Finding a basis from an equation

PS5 q2(a)

For $H = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$, solve for $z = -x - y$:

$$(x, y, z) = x(1, 0, -1) + y(0, 1, -1).$$

These two vectors span H and are independent (their first two coordinates force zero coefficients), hence

$$\boxed{\{(1, 0, -1), (0, 1, -1)\} \text{ is a basis, and } \dim H = 2.}$$

General method: solve the defining linear constraints, write the general vector as a linear combination of free-variable directions, and check independence.

04 MINIMAL NOTES

Linear maps and their matrices

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 59–74

Linearity. [3.0.1] $T : V \to W$ is linear iff PS6 q1,7

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v) \qquad (u, v \in V;\ \alpha, \beta \in \mathbb{F}).$$

In particular $T(0) = 0$. A linear map is completely determined by its values on a basis.

Matrix of a linear map. [3.2.1–3.2.5] Let $E = \{e_1, \dots, e_n\}$ be a basis of V and $F = \{f_1, \dots, f_m\}$ a basis of W. The matrix $[T]_F^E$ has columns PS6 q2–4

$$\boxed{[T]_F^E = \big([T(e_1)]_F \ \cdots \ [T(e_n)]_F\big)},$$

and satisfies

$$\boxed{[T(v)]_F = [T]_F^E [v]_E.}$$

For $T : \mathbb{F}^n \to \mathbb{F}^m$ in standard bases, this is ordinary multiplication $T(v) = Av$; the columns of A are $T(e_j)$.

Composition = multiplication. [3.2.3] If $T : \mathbb{F}^n \to \mathbb{F}^m$ has matrix B and $S : \mathbb{F}^m \to \mathbb{F}^k$ has matrix A, then PS6 q5–7

$$[S \circ T] = AB, \qquad (AB)_{ij} = \sum_{\ell=1}^{m} a_{i\ell} b_{\ell j}.$$

Thus matrix multiplication is associative and distributive, but not generally commutative. Also

$$(AB)^T = B^T A^T.$$

Invertibility. [3.3.1–3.3.4] $A : V \to W$ is invertible if there exists $A^{-1} : W \to V$ with PS8 q1–3

$$AA^{-1} = I_W, \qquad A^{-1}A = I_V.$$

The inverse is unique, and

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (A^T)^{-1} = (A^{-1})^T.$$

An invertible map takes bases to bases. Hence isomorphic finite-dimensional vector spaces have the same dimension.

The conceptual picture.

$$\boxed{\text{abstract vector} \xrightarrow{\text{choose basis}} \text{coordinate column} \xrightarrow{\text{matrix}} \text{coordinate column.}}$$

A matrix is not the map itself: it is the map written in chosen coordinates.

Fast recognition

Transformation	Linear?
rotation/reflection/scaling about 0	yes
differentiation	yes
translation $v \mapsto v + v_0$, $v_0 \neq 0$	no, because $T(0) \neq 0$
nonlinear coordinate rule such as $(x, y) \mapsto (x^2, y^2)$	no

04 WORKED METHODS

The matrix of a map: columns are images

Small integer examples; the method is the point.

A derivative acting on a polynomial space

Let $D : P_3 \to P_2$ be $D(p) = p'$. Use the *domain basis* $E = (1, t, t^2, t^3)$ and *codomain basis* $F = (1, t, t^2)$. To build the matrix: PS6 q2(c),7

1. Apply D to *each domain basis vector*:

$$D(1) = 0, \quad D(t) = 1, \quad D(t^2) = 2t, \quad D(t^3) = 3t^2.$$

2. Write each output as an F-coordinate column and put these columns *side by side*:

$$\boxed{[D]_F^E = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}}.$$

To check the method, for $p(t) = 2 + 3t + t^2 + 2t^3$:

$$[D]_F^E [p]_E = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 6 \end{pmatrix} = [3 + 2t + 6t^2]_F.$$

This is precisely $p'(t) = 3 + 2t + 6t^2$.

A map in the standard basis

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x, y) = (2x + y, 3y)$. Then PS6 q2(a)

$$T(e_1) = (2, 0), \qquad T(e_2) = (1, 3), \qquad \boxed{[T] = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}}.$$

Column rule, not row rule: the first column is $T(e_1)$ and the second is $T(e_2)$. Later we will express *the same map* in a different basis and obtain a diagonal matrix.

> **Two uses of rows and columns that must not be confused.** To *build a map matrix*, put the images of basis vectors into its **columns**. To *multiply matrices*, take a **row** of the left matrix against a **column** of the right matrix.

05 MINIMAL NOTES

Linear systems and elimination

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 75–93

Matrix form and geometry. [4.1–4.2.1] A system of m linear equations in n unknowns is PS2 q3

$$\boxed{Ax = b}, \qquad A \in M_{m,n}(\mathbb{F}),\ x \in \mathbb{F}^n,\ b \in \mathbb{F}^m.$$

Geometrically, each real equation is a hyperplane; the solution set is their intersection. Equivalently, if $A = (a_1 \ \cdots \ a_n)$ then

$$Ax = b \iff x_1 a_1 + \cdots + x_n a_n = b.$$

So solving is also asking whether b lies in the span of the columns.

Row operations and REF. [4.2.2–4.2.7] Elementary row operations preserve the solution set: swap rows, scale a row by a nonzero scalar, add a multiple of one row to another. In **echelon form**, nonzero rows lie above zero rows; the first nonzero entries (*pivots*) move strictly right as we move down. Use *back-substitution* to solve. In **reduced echelon form (REF)**, additionally each pivot is 1 and is the only nonzero entry of its column. The REF is unique; the answers can be read directly. PS7 q1–4

A column without a pivot corresponds to a **free variable**. Once free variables are chosen, pivot variables are determined.

Read everything from pivots [4.2.8–4.2.12]. For $A \in M_{m,n}(\mathbb{F})$:

- $Ax = b$ is inconsistent iff the echelon form of $(A \mid b)$ has a pivot in the last column.
- $Ax = b$ is solvable for every b iff A has a pivot in every row.
- $Ax = b$ has at most one solution iff A has a pivot in every column.
- Columns of A are independent iff pivot in every column; they span $\mathbb{F}^m$ iff pivot in every row.

For square A, all of these collapse to: pivot in every row/column $\iff$ A invertible.

Inverse by elimination. [4.2.7] For square A, row-reduce the augmented matrix PS8 q1,3

$$(A \mid I) \longrightarrow (I \mid A^{-1}).$$

If the left block cannot be reduced to I, then A is not invertible.

Principle of linearity. [4.3.1–4.3.2] For an inhomogeneous equation $Ax = b$, suppose x_0 is one particular solution and let PS2 q5

$$H = \{x : Ax = 0\} = \operatorname{Ker} A.$$

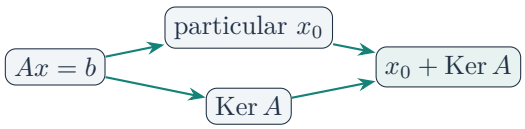
Then

$$\boxed{\{\text{all solutions of } Ax = b\} = x_0 + H.}$$

In words:

$$\text{general solution} = \text{one particular solution} + \text{general homogeneous solution}.$$

Thus every consistent linear system has an affine solution set: a translated subspace.



05 WORKED METHODS

Elimination, REF, free variables and inverse

Small integer examples; the method is the point.

Solving a system by row operations

PS7 q2; PS2 q3

Solve $x+y+z=4$, $x-y+z=2$, $y-z=0$. Put the coefficients and right-hand side in an *augmented matrix*:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & -1 & 1 & 2 \\ 0 & 1 & -1 & 0 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -2 & 0 & -2 \\ 0 & 1 & -1 & 0 \end{array}\right).$$

$$\xrightarrow{R_2 \leftarrow -R_2/2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 \end{array}\right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{array}\right).$$

Turn the last pivot into 1 and clear above the pivots:

$$\xrightarrow{R_3 \leftarrow -R_3,\ R_1 \leftarrow R_1 - R_2 - R_3} \boxed{\left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right)}.$$

Thus the unique solution is $\boxed{(x,y,z)=(2,1,1)}$. The last matrix is in *reduced echelon form (REF)*: every pivot is a 1 with zeros above and below.

Three possible outcomes

PS7 q3–4

- $[\,1\ 1 \mid 2\,;\ 0\ 0 \mid 1\,]$: impossible row $0=1$, so **no solutions**.
- $x+y+z=2$: choose $y=s, z=t$ freely; $x=2-s-t$, hence **infinitely many solutions**.
- Three pivots for three unknowns, as above: **one solution** (provided consistent).

Finding an inverse using $[A \mid I]$

PS8 q1

For $A=\left(\begin{smallmatrix}1&1\\1&2\end{smallmatrix}\right)$, reduce both blocks *together*:

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 1 \end{array}\right).$$

Therefore $\boxed{A^{-1}=\left(\begin{smallmatrix}2&-1\\-1&1\end{smallmatrix}\right)}$. Check $AA^{-1}=I$. If the left side *cannot* become I, no inverse exists.

06 MINIMAL NOTES

Image, kernel, rank and change of basis

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 94–104

Image and kernel. [4.4.1–4.4.6] For $A : V \to W$, PS9 q1,4

$$\operatorname{Im} A = \{Av : v \in V\} \subseteq W, \qquad \operatorname{Ker} A = \{v \in V : Av = 0\} \subseteq V.$$

Both are subspaces. For a matrix $A = (a_1 \ \cdots \ a_n)$,

$$\operatorname{Im} A = \operatorname{span}\{a_1, \ldots, a_n\}.$$

Furthermore

$$A \text{ surjective} \iff \operatorname{Im} A = W, \qquad A \text{ injective} \iff \operatorname{Ker} A = \{0\}.$$

Rank-nullity. [4.4.7–4.4.10] PS9 q1–3

$$\operatorname{rank} A = \dim \operatorname{Im} A, \qquad \operatorname{nullity} A = \dim \operatorname{Ker} A.$$

If $A : V \to W$ and V is finite-dimensional,

$$\boxed{\operatorname{rank} A + \operatorname{nullity} A = \dim V.}$$

For a matrix in echelon form:

- rank = number of pivots;
- nullity = number of free columns;
- the *original* columns of A corresponding to pivot columns form a basis of $\operatorname{Im} A$;
- solve $Ax = 0$ to get a basis of $\operatorname{Ker} A$.

Also $\operatorname{rank} A = \operatorname{rank} A^T$. If $\dim V = \dim W = n$, injective $\iff$ surjective $\iff$ rank n $\iff$ nullity 0 $\iff$ invertible.

Change of coordinates. [4.5] For bases E, F of the same space, the transition matrix $[I]_F^E$ converts E-coordinates into F-coordinates: PS9 q6–7

$$\boxed{[v]_F = [I]_F^E [v]_E}, \qquad [I]_E^F = ([I]_F^E)^{-1}.$$

Its jth column is $[e_j]_F$.

If $T : V \to V$, then changing from basis E to F changes its matrix by

$$\boxed{[T]_F^F = P^{-1}[T]_E^E P},$$

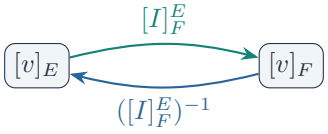
where $P = [I]_E^F$ converts F-coordinates *into* E-coordinates. Read the direction: $[v]_E = P[v]_F$.

Similarity. [4.5.5–4.5.6] Square matrices A, B are similar if PS9 q7

$$A = P^{-1}BP$$

for some invertible P. Conceptually:

$$\boxed{A \sim B \iff A \text{ and } B \text{ represent the same linear map in different bases.}}$$



Keep in mind. Row operations preserve the solution set of a system, but they do *not* preserve the column space as a literal subset. To find a basis of $\operatorname{Im} A$, use pivot positions from the echelon form and take the corresponding columns of the original A.

06 WORKED METHODS

Kernel, column space and changing basis

Small integer examples; the method is the point.

Finding $\operatorname{Ker} A$, $\operatorname{Im} A$, rank and nullity

PS9 q1

Take

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} : \mathbb{R}^3 \to \mathbb{R}^2.$$

Step 1: kernel. Solve $Ax = 0$:

$$x_1 + x_3 = 0, \quad x_2 + x_3 = 0 \implies (x_1, x_2, x_3) = t(-1, -1, 1).$$

So $\boxed{\operatorname{Ker} A = \operatorname{span}\{(-1,-1,1)\}}$ and nullity $A = 1$.

Step 2: image. The first two columns $(1,0)^T, (0,1)^T$ are independent; the third equals their sum. Hence

$$\boxed{\operatorname{Im} A = \operatorname{span}\{(1,0),(0,1)\} = \mathbb{R}^2}, \quad \operatorname{rank} A = 2, \quad 2 + 1 = \dim \mathbb{R}^3.$$

In larger problems: row-reduce A to locate pivot columns, then take those columns from the *original* A as an image basis. Solve the homogeneous system for the kernel.

Converting coordinates

PS9 q6–7

Let $E = (e_1', e_2')$ with $e_1' = (1,0)$ and $e_2' = (1,1)$. A vector v has standard coordinates $[v]_S = (3,2)^T$. The basis matrix is

$$P = \begin{pmatrix} e_1' & e_2' \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}.$$

Columns of P are the *new basis vectors in old coordinates*. Thus

$$[v]_S = P[v]_E, \qquad \boxed{[v]_E = P^{-1}[v]_S = \begin{pmatrix} 1 \\ 2 \end{pmatrix}}.$$

Indeed $v = 1(1,0) + 2(1,1) = (3,2)$. Always check by recombining the basis vectors.

Same linear map, different matrix

PS9 q7(b)

The map from Section 04 has standard matrix

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}.$$

Since $Ae_1' = 2e_1'$ and $Ae_2' = 3e_2'$, its matrix in basis E is

$$\boxed{[T]_E = P^{-1}AP = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}}.$$

Procedure: E-coordinates $\xrightarrow{P}$ standard coordinates $\xrightarrow{A}$ standard output $\xrightarrow{P^{-1}}$ E-output. Hence $P^{-1}AP$, not PAP^{-1}.

07 MINIMAL NOTES

Determinants

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 105–120

Meaning and the 2×2 case. [5.0–5.1] For PS10 q1

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \det A = ad - bc.$$

For a real 2×2 matrix, $|\det A|$ is the area scale factor; the sign records orientation. In n dimensions, determinant is the signed volume scale factor.

Three defining properties. [5.2.1–5.2.4] As a function of the columns, det is:

- multilinear (linear in each column separately);
- alternating: swapping two columns multiplies it by -1;
- normalized: $\det I = 1$.

These properties uniquely determine the determinant.

Rules worth memorising. [5.2.5–5.2.15] PS10 q2–6

Operation / property	Effect
swap two rows/columns	multiply determinant by -1
scale one row/column by α	multiply determinant by α
add multiple of one row/column to another	unchanged
triangular/diagonal matrix	product of diagonal entries
transpose	$\det(A^T) = \det A$
product	$\det(AB) = \det A \det B$
invertibility	A invertible $\iff \det A \neq 0$

Keep in mind. $\det(A + B)$ is generally *not* $\det A + \det B$, and $\det(\alpha A) = \alpha^n \det A$ for $n \times n$ matrices.

Cofactors. [5.2.16–5.2.20] Delete row i and column j to form the minor $M_{ij}(A)$; the cofactor is PS10 q1(a)

$$C_{ij}(A) = (-1)^{i+j} M_{ij}(A).$$

Expanding along any row i or column j:

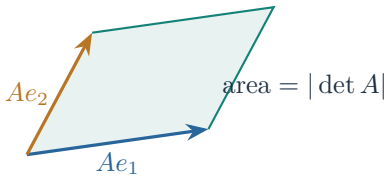
$$\det A = \sum_{k=1}^{n} a_{ik} C_{ik}(A) = \sum_{k=1}^{n} a_{kj} C_{kj}(A).$$

Choose a row/column with many zeros.

Adjugate inverse formula. [5.3.1] Let $C = (C_{ij}(A))$. If $\det A \neq 0$, then PS10 q1(a)

$$A^{-1} = \frac{1}{\det A} C^T.$$

For computation, elimination is usually quicker; the cofactor formula is structurally important.



07 WORKED METHODS

Computing determinants in two reliable ways

Small integer examples; the method is the point.

A 3×3 determinant by cofactor expansion

PS10 q1(a)

Choose a row with many zero entries:

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix}.$$

Across row 1, the cofactor signs are $+, -, +$, so

$$\begin{aligned} \det A &= 1\begin{vmatrix} 3 & 1 \\ 1 & 1 \end{vmatrix} - 2\begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} + 0\begin{vmatrix} 0 & 3 \\ 2 & 1 \end{vmatrix} \\ &= 1(3-1) - 2(0-2) = \boxed{6}. \end{aligned}$$

Method: cross out row i and column j, compute the 2×2 determinant, multiply by $(-1)^{i+j}a_{ij}$ and add. Expansion works along *any* row or column.

The same determinant by row operations

PS10 q1(b)

Adding a multiple of another row leaves det unchanged:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & -3 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Now upper triangular, so $\boxed{\det A = 1 \cdot 3 \cdot 2 = 6}$. No fractions are needed.

Invertibility test from det

PS10 q1

For this A, $\det A = 6 \neq 0$, hence A is invertible. If $\det A = 0$, the columns are dependent and the matrix has no inverse. For 2×2,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad (ad - bc \neq 0).$$

Choice of technique: cofactor expansion is best with zeros; row operations are generally best for bigger matrices. Track row swaps and rescalings if you use them.

08 MINIMAL NOTES

Eigenvalues, canonical forms and linear ODEs

Source: Nazar Miheisi, 4CCM113A (2025–26), pp. 121–132

Eigenvalues and eigenvectors. [6.1.1–6.1.2] For $A : \mathbb{F}^n \to \mathbb{F}^n$, a nonzero x is an eigenvector with eigenvalue λ if PS11 q1

$$Ax = \lambda x.$$

Equivalently,

$$(A - \lambda I)x = 0, \qquad \lambda \text{ eigenvalue} \iff \boxed{\det(A - \lambda I) = 0}.$$

The characteristic polynomial is $p(\lambda) = \det(A - \lambda I)$. Once λ is known, its eigenspace is

$$E_\lambda = \operatorname{Ker}(A - \lambda I),$$

and the eigenvectors are its nonzero elements.

Diagonalisation. [6.2] A is diagonalizable over $\mathbb{F}$ iff it is similar to a diagonal matrix: PS11 q2–5

$$P^{-1}AP = D.$$

Equivalently, A has a basis of eigenvectors. When $x_1, \dots, x_n$ are independent eigenvectors with eigenvalues $\lambda_1, \dots, \lambda_n$,

$$P = (x_1 \ \cdots \ x_n), \qquad D = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

Distinct eigenvalues give linearly independent eigenvectors.

2×2 **canonical forms** [6.2.1–6.2.2]. Over $\mathbb{C}$, every 2×2 matrix is similar to exactly one type

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

For a real 2×2 matrix, a complex-conjugate pair $\alpha \pm i\beta$ ($\beta \neq 0$) corresponds over $\mathbb{R}$ to

$$\begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}.$$

A repeated real eigenvalue is diagonalizable iff its eigenspace has dimension 2.

Systems of ODEs. [6.3] For the homogeneous constant-coefficient system PS11 q6

$$\frac{du}{dt} = Au,$$

if $Ax = \lambda x$ then

$$u(t) = e^{\lambda t}x$$

is a solution. If A is diagonalizable with an eigenbasis $x_1, \dots, x_n$, then the general solution is

$$\boxed{u(t) = \sum_{k=1}^{n} C_k e^{\lambda_k t} x_k.}$$

Initial data $u(0) = u_0$ determines the constants by expressing u_0 in the eigenbasis.

One course in one chain. []

vectors → span/subspaces → bases/dimension → linear maps
→ matrices → elimination → rank/kernel → change of basis
→ det → eigenvalues → simple canonical forms/ODEs.

Most of the course is the same object viewed from three angles: **linear combinations**, **linear maps**, and **coordinates**.

08 WORKED METHODS

Eigenvalues, eigenvectors and a simple ODE

Small integer examples; the method is the point.

Find the eigenvalues: solve one scalar equation

PS11 q1

Re-use $A = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right)$. Compute

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 0 & 3-\lambda \end{vmatrix} = (2-\lambda)(3-\lambda) = 0.$$

Thus the eigenvalues are $\boxed{\lambda = 2 \text{ and } \lambda = 3}$.

Find the eigenvectors: solve one homogeneous system for each λ

PS11 q1–2

For $\lambda = 2$,

$$(A - 2I)x = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \iff y = 0; \quad E_2 = \operatorname{span}\{(1,0)\}.$$

For $\lambda = 3$,

$$(A - 3I)x = \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \iff x = y; \quad E_3 = \operatorname{span}\{(1,1)\}.$$

The eigenvectors $(1,0)$ and $(1,1)$ are independent, so with

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad \boxed{P^{-1}AP = D}.$$

This is *exactly* the change-of-basis computation from Section 06. Eigenvectors are a basis in which the map becomes simple diagonal scaling.

When diagonalisation fails

PS11 q2–3

$B = \left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ has only eigenvalue 2, with $E_2 = \operatorname{span}\{(1,0)\}$. One eigenvector direction cannot form a basis of $\mathbb{R}^2$: B is **not diagonalizable**.

Solving a constant-coefficient system

PS11 q6

For $u'(t) = Au(t)$ with our diagonalizable A, the eigenvectors give the general solution

$$\boxed{u(t) = c_1 e^{2t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.}$$

If $u(0) = (2,1)^T$, then $c_1 + c_2 = 2$, $c_2 = 1$, so $c_1 = c_2 = 1$ and

$$\boxed{u(t) = \begin{pmatrix} e^{2t} + e^{3t} \\ e^{3t} \end{pmatrix}.}$$

Recipe: find eigenvalues and eigenvectors; form an independent eigenbasis; take $e^{\lambda t}$ times each eigenvector; choose constants using $u(0)$.